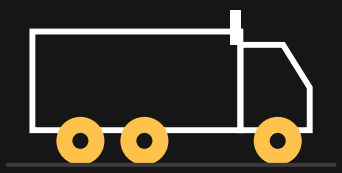
DRIVER'S MANUAL

TRUCKHUB



A live instrument cluster for American Truck Simulator & Euro Truck Simulator 2

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TruckHub reads live telemetry from ATS/ETS2's own official SDK - nothing here is a guess or a simulation of a simulation. Every gauge, color, and label in this manual traces back to a real value the game itself is reporting at that instant.

SECTION 01

Getting Started

TruckHub is a companion instrument cluster that runs alongside American Truck Simulator or Euro Truck Simulator 2. It doesn't mod the game and doesn't read your screen - it connects to the same official telemetry SDK plenty of racing-sim and trucking dashboards already use, and draws what it gets from that feed as a second, always-on-top card next to your game window.

Launch TruckHub before or after the game - order doesn't matter. The card sits idle with a single line reading *"Waiting for ETS2 / ATS to start…"* until a supported game is actually running, then fills in on its own. Close the game and it goes back to waiting rather than freezing on stale numbers.

Installing & first run

Run **TruckHub.exe** - there is nothing else to install. At startup TruckHub looks for your Euro Truck Simulator 2 and American Truck Simulator installs through Steam and copies its own small telemetry plugin, **scs_telemetry_hub.dll**, into each game's `bin\win_x64\plugins` folder. A game only loads plugins when it starts, so if TruckHub put the plugin in while that game was already running, **restart the game once**. After that it is automatic every time.

If a game isn't found (a non-Steam copy, or a game library on a drive Steam can't see), open Settings and point TruckHub at the game's folder yourself - see *Game install locations* in Section 07.

That plugin is the **only** file TruckHub puts in your game folders. The older third-party `scs-telemetry.dll` plugin is no longer part of TruckHub at all (nothing reads it), and a copy already sitting in your plugins folder is left exactly as it is. If another tool has installed its own plugin and named the file in a manifest (a `.json` file beside it), TruckHub leaves that file alone too and notes it in its log.

The plugin is built on the same official telemetry SDK as before and also covers ATS's **Road Trip DLC** "Quick Job" personal-car courier jobs: job details, Trip Needs and Economy all work there the same as on regular freight.

> **TIP** In the game, turn on **Options > Gameplay > HUD > Route Advisor: speed limit**. That is the setting that feeds the speed limit sign - without it the sign stays on its grey placeholder.

The card itself

Click and drag anywhere on the card's background to move it. The bottom-right corner is a resize handle - drag it to scale the whole card, fonts and all, up or down. TruckHub remembers where you left the card and how big you made it, and puts it back there next time. If you turn a row off in Settings the card simply gets shorter; turn it back on and it grows again.

Three tabs sit to the right of the card - **GPS**, **LOG**, and **EXT**. Each opens something separate: a live road map, a running job history, and a second window of instruments the main card has no room for. All three are covered in their own sections below.

TIP TruckHub stays on top of other windows by design, so you can glance at it without alt-tabbing. If it's in the way, resize it smaller or drag it to a corner - there's no "minimize to tray" here, it's meant to always be visible while you drive.

Under the hood: CoreBalancer

TruckHub quietly measures real per-core CPU load every minute and steers its own process - and the GPS map's rendering subprocess - away from whichever cores the game is currently loading hardest, instead of guessing a fixed core to avoid. A static "always reserve core 0"-style rule was tried first and, once actually measured against live per-core load, turned out not to match how load really moves around on a real machine while ATS or ETS2 is running.

It also decides how hard the GPS map itself is allowed to work: the map renders at 60fps when the system has headroom, and drops to a steadier 30fps on its own if overall load is high, so a demanding moment in-game - or something else entirely running on your PC - doesn't have to fight the map for the same CPU and GPU time. Changes are deliberately unhurried: a core reassignment only happens when the busiest cores are clearly worse than what TruckHub is already avoiding, and never more than once every five minutes, so ordinary noise in the readings doesn't cause constant switching. None of this needs any configuration - it runs by itself the entire time TruckHub is open.

SECTION 02

The Main Dashboard

Top to bottom, exactly the order it appears on the card.

Title bar

Mod Order

Opens a suggested load order for your installed mods, built from community-recommended category tiers. It's a reference list only - TruckHub never touches your Mod Manager, you still reorder mods yourself. See Section 08.

Settings

Opens the two-tab Settings window (Section 07). Glows amber whenever a TruckHub update is available - see the update note in Section 07.

Fullscreen

Toggles the card between its normal size and a larger fullscreen layout.

Close

Closes TruckHub. The game keeps running - this only closes the HUD.

Speed limit sign & turn signals

The sign matches whichever game is running: American Truck Simulator gets the familiar white rectangular **SPEED LIMIT** sign, Euro Truck Simulator 2 gets the red-ringed white circle used across Europe. With no speed limit data for the current road, a dim grey placeholder shows instead of a guessed number (if it *never* shows a number, check the Route Advisor setting in Section 01).

Two optional sounds go with it: a beep when the posted limit changes (higher tone for up, lower for down) and a double beep when your speed turns red - see *Speed alerts* in Section 07.

The two arrows either side of the sign are your actual turn signals, driven straight off the game's own bulb state - they blink at the real rate, and flash together for hazards, because hazards are the same two bulbs turned on at once.

Lights & controls row

Nine icons, each dim grey when off and lit in its real dashboard color the instant it's active - here's what each one actually is, so you know it on sight the first time it lights up.

P	**Parking brake**	Lit white while the parking brake is set. Off = brake released.
R	**Retarder**	Lit green while engaged - the ring's letter switches to the current level number (1, 2, 3…) instead of staying a plain "R". Only shown on trucks that actually have a retarder fitted.
J	**Engine brake**	Lit green while engaged. Labeled "J" (Jake brake) in ATS, "E" (engine brake) in ETS2 - same feature, the letter just matches what that game's audience calls it.

	**Sidelights**	A plain lamp, no beam lines. Lit green while your sidelights/parking lights are on.
	**Low beam**	Lamp with three beams angled downward. Lit green while low beams are on.
	**High beam**	Lamp with three straight, level beams. Lit blue while high beams are on - the one deliberate exception to the green/amber/red rule (see Section 03).
	**Beacon**	A domed light with rays. Lit amber while your roof beacon/hazard light bar is on.
	**Differential lock**	A padlock. Lit amber while the diff lock is engaged - it changes how the truck handles, so it's flagged even though nothing's wrong.
	**Lift / tag axle**	A wheel with an arrow above it. Lit amber while the axle is raised; dim while it's down (normal). Only appears at all on a truck that actually has one - see Section 03 for what LIFT/TAG mean on the EXT window's own axle readout.

Warning cluster

Seven telltales mirroring the exact warning lights on the game's own dashboard - TruckHub adds no thresholds of its own here, if one of these lights up, the game itself decided it should. Every light in this row goes dark if the truck's electrics are switched off, the same way a real dashboard goes blank with the ignition off.

	**Low air pressure**	The real SAE/ISO air-brake warning symbol - a circled exclamation mark. Red once the air tank pressure drops too low to brake reliably.
	**Emergency / parking brake**	The same ring-and-"P" shape as the parking brake status icon above, but this one lives in the warning row and turns red specifically when critically low air pressure has auto-applied the brake for you - not a brake you set yourself.
	**Oil pressure**	An oil can with a drip. Red once engine oil pressure drops below safe.
	**Engine temperature**	A thermometer. Red once coolant temperature runs too hot.
	**Low fuel**	A fuel pump. Amber once your tank drops below the game's own low-fuel threshold.
	**Low AdBlue**	A droplet. Amber once your AdBlue/DEF level runs low.
	**Battery / charging**	A battery with its two terminals. Red if the charging system fails or voltage drops critically - on a real truck this can mean losing electrical power entirely, which is why it's red rather than amber.

Speed, RPM, and the odometer

Both dials sweep like a real analog gauge rather than snapping to a new position. Speed's needle and digital readout turn green at or under the posted limit, amber up to 5 over, and red beyond that - there's no fixed color band painted on the dial face itself, since what counts as "safe" changes with every speed limit sign you pass. RPM works the other way: its dial has a fixed red zone painted onto the last stretch of the arc, because that position is always correct once the dial is scaled to your engine's own redline, whatever RPM number that happens to be.

Directly under the speedo sits a six-digit mechanical-style odometer, ticking up in whole miles or kilometers as you drive - it flashes briefly each time a digit actually rolls over.

Fuel & AdBlue

One gauge face, two readings. Fuel is a swept needle with a painted red reserve zone in the last eighth of the dial and its own low-fuel telltale near “E”. AdBlue shares the same face as four discrete quarter-segments below the needle's pivot - the same segmented-bar convention real trucks use for DEF gauges - lighting up as soon as the level reaches into each new quarter, so it never looks empty until it actually is. With the truck's electrics off the fuel needle rests at E, like a real dash.

Cruise control & gear

Cruise shows its set speed at all times, colored green while engaged and dim grey while off. The gear box shows D/N/P/R for an automatic; for a manual truck it shows the real gear number, with an L/H suffix on trucks that have a range switch. On an H-shifter gearbox TruckHub hasn't been calibrated for yet, it falls back to showing the raw shifter slot (e.g. **S4**) rather than a confidently wrong gear number - see Gearbox Calibration in Section 07.

Job panel

While a job's active you see the route (pickup to drop-off, with company names), the cargo and its weight, then three boxes side by side:

Payout - what the job pays, with the pay per mile (or per kilometer) under it.

Distance - the big number is how far you have actually driven on this job, counting **up** from zero; the smaller *of N* beneath it is the job's total length. A ferry crossing, a detour or a rerouted trip can make the real distance different from what the job board quoted, and this stays accurate either way.

Time left - how long until the deadline in game time. It falls as the game clock advances, jumps when you rest, and reads **Late** followed by the overshoot once the deadline has passed.

The Distance total is remembered while the job is in progress: close TruckHub, or quit and reload the game half-way through a long haul, and it still shows the job's full length rather than what's left. It is forgotten when the job is delivered or cancelled. For a job TruckHub first sees part-way through, it uses the job's quoted length, which can differ a little from the in-game GPS length. Until your cargo is loaded the box counts the way to the pickup, then switches to the full route.

Below that: **Trip Needs**, the percentage of your current fuel this job will actually take to finish (red if you're short), and **Economy**, your rolling fuel consumption in MPG or L/100km depending on your unit settings.

Rest timer

Green with over an hour left, amber under an hour, red at or past due. This row only appears if you've told TruckHub you have fatigue simulation switched on in-game (Section 07) - the game reports the same placeholder countdown whether the simulation is on or simply hasn't started ticking yet, so TruckHub has no way to tell those two states apart on its own.

SECTION 03

Color & Symbol Legend

One page, for whenever a light comes on and you can't remember what it means.

Green	Normal, safe, or a driving aid working as intended (cruise, retarder, engine brake, turn signals, low beam, sidelights).
Amber	Needs attention soon, or a deliberate state that changes handling (diff lock, lift axle up, low fuel/AdBlue, beacon, near-due rest timer).
Red	Safety-critical fault or danger: low air pressure, oil pressure, overheating, dead battery/charging failure, over the speed limit by more than 5, an axle unexpectedly off the ground.
Blue	High beam, specifically - the one near-universal dashboard exception to the green/amber/red scheme.
Dim grey	Off, disengaged, or not applicable right now.
White	A neutral reading with no state to flag - a plain number or label.

Why the colors sometimes mean different things

A couple of icons bend the rule above on purpose, matching how the real vehicle they represent actually behaves: the retarder and engine brake glow **green** when engaged, not amber, because using them is a deliberate braking choice working exactly as intended - the same category as cruise control, not a fault. The differential lock and lift axle glow **amber** when active even though nothing is wrong, because both genuinely change how the truck handles and are worth being aware of while they're on.

Suspension / axle labels (EXT window)

W1, W2, W3…	An ordinary steer or drive wheel - just numbered in order.
LIFT	A liftable axle (a “tag” or “pusher”, depending where it sits) - read directly from the game's own axle data, not guessed by position.
TAG	A fixed auxiliary axle - neither driven, steered, nor liftable - carrying extra load but never raised.
UP **(amber)**	A lift axle currently raised. Its suspension bar goes empty on purpose - there's nothing meaningful to read with the wheel off the ground.

SECTION 04

The GPS Map

Opened from the **GPS** tab. This is a real reconstruction of whichever game is running - the full ATS road network, or ETS2's, including the UK (reachable in-game only by ferry, and included on the map as real, drivable roads even though nothing connects it to the continent by land) - not a generic map with your position pasted on top. Your truck's position and heading update live, and once a job's destination is known, TruckHub computes a route that actually follows real roads to get there.

The route is calculated when the job starts (or when you open the map with one already active), and the part you've already driven is trimmed away behind you so you only see what is ahead. It is not rebuilt behind your back when you leave it: you decide when to reroute - see *When you leave the route* below.

This includes ferry and train crossings (the UK to mainland Europe, most notably) - the route carries straight across the water as part of the same job, the same way it treats any other stretch of the trip.

TruckHub's GPS has no connection to the in-game GPS - each works out its own route independently, so on a long trip they can occasionally choose different roads. That is a limit of what the games' SDK exposes, not a fault in either one.

First time only: after you first install TruckHub, or after an update that changes the map's own data, TruckHub unpacks the map's files in the background. That is a few minutes of heavy processor use, and the map may be slow to appear until it finishes. Ordinary updates that don't touch the map data don't repeat it.

On the map

Highway shields

Interstate, US Route, and State Route signs drawn in their real MUTCD shapes and colors, in ATS. ETS2 gets real European road signs too - green plates for E-roads, national motorways, and expressways; blue for national and regional roads - matching how the real signs look rather than a US-style shape.

City & town names

Sized and brightened by importance - major cities render larger and whiter, small towns smaller and dimmer.

State / country names

Muted, letter-spaced capitals in the background - context, not something competing for attention with cities.

Facility markers

Real pictograms rather than letters - see below.

Company names & logos

Every delivery company's marker shows its real in-game name and logo, sourced directly from the game's own files - both ATS and ETS2.

Facility markers

Fuel	A pump icon - only genuine, truck-usable gas stations. Decorative AI-traffic-only stations are filtered out.
Service	A wrench icon, for repair shops and truck dealers.
Rest stop	A bed icon, for parking areas you can actually stop and sleep at.
Weigh station	A plain “W” badge.

The map window is fully independent and resizable, with its own minimize, maximize, and reset-to-default-size controls in the title bar.

The toolbar

A row of buttons in the map's bottom-left corner. A button that is lit amber is switched on.

Follow	Press to centre the map on your truck and keep it centred as you drive; press again to turn it off so you can drag the map around and look ahead. It starts **off** when the map opens.
Pin	Press Pin, then click anywhere on the map: TruckHub drops a green pin there and draws a route to it from where your truck is. Use it for a fuel stop, a rest area, or a trip with no job active. One press places one pin - press Pin again for another.
Clear	Removes your pin and goes back to the current job's route (or to no route, if you have no job).
LAN	Shares the live map with other devices on your network - see LAN Mode below.
Turns	Shows or hides the next-turn sign (below). On by default.
Reroute	Not always there: it appears only while you are a mile (1.6 km) or more off the route. See *When you leave the route* below.

Next-turn sign

A sign in the map's bottom-right corner while a route is active, showing the next turn's direction and distance - green in ATS, European blue in ETS2. The distance is in the game's own miles (or kilometres), the way your odometer and the in-game GPS count them: TruckHub measures how far the road you are on stretches your odometer, so it tracks closely along a steady road and can be a little off for a turn beyond a change of road type. The sign appears once a turn is about two miles of driving ahead. The **Turns** button switches it off.

When you leave the route

Earlier versions rebuilt the route as soon as you strayed 200 m (650 ft) from it, so a fuel stop down an access road forced a recalculation. Now **you** decide when it is rebuilt.

> **NOTE** **Miles of driving vs odometer miles.** The games squeeze their maps and run the game clock fast, so the odometer, the in-game GPS and TruckHub's Distance box count several miles for each mile the truck physically travels: about 3 on slow roads and in towns, about 15 on UK motorways, and a steady 16 in a multiplayer convoy. TruckHub measures distance from the route in the game world itself, so a "mile" here is a mile of actual driving - about a minute at highway speed - which your odometer counts as anywhere from about 3 miles on slow roads to 15 or more on motorways.

Within a mile	Nothing happens. Pull in for fuel, park at a rest area, take an exit to a station: the route line stays exactly where it was. The next-turn arrow is hidden while you are away from the route (it would point at a turn you are no longer approaching) and comes back by itself when you rejoin.
A mile (1.6 km) or more	A red **OFF ROUTE** warning appears at the top centre of the map, and a red **Reroute** button appears at the left of the toolbar. The button is only there while you are that far off.
Press Reroute	TruckHub recomputes the route from where the truck is now to the same destination. The warning and the button disappear and the new route is drawn.
Back on the route	The warning clears by itself once you are within 200 m (650 ft) of the route again. It does not flicker on and off while you wander between 200 m and a mile on the way back.

The Reroute hotkey: Ctrl+Alt+F8 does the same thing from anywhere, including while the game has focus, so you never have to alt-tab to the map. It also works when the warning isn't showing (it simply recomputes from your current position), and does nothing, harmlessly, if the map hasn't been opened this session or there is no destination. Neither ATS nor ETS2 binds F8 to anything, so pressing it can never also trigger a game action. If another program on your PC already uses Ctrl+Alt+F8, TruckHub notes it in its log and the Reroute button on the map still works.

Why a mile? The in-game GPS doesn't recalculate the instant you leave the road either - it gives you a little leeway to make a U-turn. A mile of driving is far enough that an access road to a rest area or station never triggers anything, but close enough that a genuine wrong turn is flagged while it is still easy to correct.

Prefer the old behaviour? **Settings > App Configure > GPS route** has *Recompute the route automatically when I leave it* (off by default): with it on, TruckHub rebuilds the route for you beyond 200 m, with no warning and no button. Phones and tablets using LAN Mode don't get the OFF ROUTE warning or the Reroute button - those exist only on the map window on your PC.

LAN Mode

A **LAN** button sits in the map's own toolbar, alongside Pin and Clear. Turn it on and this same live map - your position, heading, and route - becomes viewable from another device on your home network through a plain browser: a phone or tablet propped up on the dash, for instance. Nothing needs installing on that other device.

Turning it on asks you to choose **No SSL** or **Use SSL**. Pick No SSL unless you have a specific reason not to - it's the recommended, works-everywhere option, and the only one that's reliable on an iPhone or iPad (Safari's TLS stack doesn't get along with the self-signed certificate the SSL option needs here). Once it's on, the address to type into the other device's browser is shown right there in the corner of the map.

> **NOTE** The remote view is read-only and local-network-only. You can watch the live position and route from another device, but placing or clearing a pin, and turning LAN Mode itself on or off, only ever happens from the PC's own map window - and it only reaches devices on the same network, not the internet.

SECTION 05

Extended Instruments (EXT)

Everything the main dashboard doesn't have room for.

Opened from the **EXT** tab. Same rule as the main card: nothing shows until a supported game connects. Every element here can be switched on or off individually from this window's own settings button, separately from the main dashboard's toggles.

The six gauges

Oil Pressure	0–100 psi.
Oil Temperature	Green / amber / red as it climbs toward the high end of the dial.
Water Temperature	Red once the game's own overheat warning trips.
Battery	Scaled to the game's electrical system: 10-16 V in American Truck Simulator (12 V trucks) and 18-30 V in Euro Truck Simulator 2 (24 V trucks). The number is the game's own voltage reading, and the needle sits at the right place on the dial for whichever game you're in.
Brake Air Pressure	0–150 psi, red once the game's own low-air warning trips.
Brake Temperature	Green under 260°C, amber 260–343°C, red above 343°C - real heavy-truck drum-brake service limits. A long descent can genuinely run brakes into the amber zone; sustained heat past the red line risks fade and drum damage.

Suspension deflection

One segmented bar per wheel, truck and trailer shown separately - filling from the bottom in six steps, the same segmented-cell look as the main dashboard's AdBlue gauge. It's the closest reading the game exposes to “how loaded is this axle” (more sag reads as more segments lit), not a calibrated weight scale.

Wheels are only ever shown for axles your specific truck or trailer actually has - there's no padding out to a generic maximum wheel count. See Section 03 for what the LIFT / TAG / UP labels mean.

Trailer axle status

Per-wheel state for the attached trailer: **DOWN** (on the ground, normal), **LIFTED** (deliberately raised), or **AIRBORNE** (unexpectedly off the ground - worth a look). Beneath the wheel list, the tractor's own lift-axle control lamp shows whether you've commanded the trailer's axle up or down from the cab.

Accessory wear

Engine, transmission, cabin, chassis, and wheel wear for the truck; body, cargo, chassis, and wheel wear for the trailer. Green under 50% worn, amber 50–80%, red beyond that.

SECTION 06

Job Log

Opened from the **LOG** tab, which drops down a drawer beside the main card. Click the pop-out button in its corner to move the log into its own separate, resizable window if you'd rather keep it away from the main dashboard.

Every job you deliver is added to the top of the list with its pickup, drop-off, payout, and distance, alongside a running **Total Payout** and **Total Mileage** for the session. The distance logged is the job's whole route, including any part you drove before closing and reopening TruckHub or reloading the game. A cancelled or failed job is simply dropped, not logged.

NOTE The log is a running tally for this session only - it isn't saved to disk, and starts empty again the next time TruckHub opens.

SECTION 07

Settings

Two tabs: App Configure and UI Setup.

App Configure

Updates

TruckHub quietly checks GitHub for a newer release each time it starts; if you're up to date or offline nothing happens and you're never interrupted. You can also press **Check for Updates** at the top of Settings at any time. A found update shows a **Download update** button here and puts a small amber glow on the Settings button on the main dashboard until you've opened Settings and seen it.

Game install locations & mod folder locations

TruckHub auto-detects standard Steam installs for both games. If a copy isn't found (a non-Steam install, or a library drive Steam isn't currently seeing), press **Browse...** under that game and choose its folder; **Clear override** goes back to auto-detection. A folder that doesn't look like a real game install is flagged and ignored rather than overriding a good auto-detected one. The same pair of buttons under *Mod folder locations* sets the folder the Mod Order tool (Section 08) scans, for anyone whose mods aren't in the usual Documents location.

Gearbox calibration

American Truck Simulator resets an H-shifter to neutral the instant the game window loses focus - so tabbing out to read the current gear doesn't work, the game's own telemetry genuinely reports neutral the moment you do it. The workaround:

1. While driving, with the game still focused, shift into the gear you want to label.
2. Press **Ctrl+Alt+G** to capture that shifter position.
3. Come back to Settings (whenever's convenient - the capture is remembered) and type in which gear that actually was, then press Set gear.

Low/High range is detected and labeled automatically once you've calibrated a position. A splitter that reuses the same physical stick slots for different gears needs its own separate capture and Set, using the splitter-specific row that appears once TruckHub detects your gearbox has one. The calibration is remembered per truck/gearbox, so switching between a few regular trucks doesn't mean starting over; **Clear calibration** wipes the current gearbox's entries if you want to redo it. (Driving an automatic? None of this applies - TruckHub shows D/N/P/R.)

GPS route

One checkbox: **Recompute the route automatically when I leave it**. **Off by default.** Turn it on to get the earlier behaviour back - as soon as you are more than 200 m (650 ft) off the route TruckHub rebuilds it for you, with no button and no warning. Choose it if you'd rather never touch a button and don't mind a recalculation whenever you stop off the highway (every fuel stop or rest area reached by a side road will trigger one). It takes effect within a few seconds; no restart needed. With it off, you get the OFF ROUTE warning, the Reroute button and the Ctrl+Alt+F8 hotkey described in Section 04.

You often stop for fuel or rest and want the route left alone	Leave it **off** (default)
You rarely stop off-route and want fully hands-off navigation	Turn it **on**
You take a wrong turn	Off: wait for the warning (a mile) or press Reroute / Ctrl+Alt+F8 at once. On: automatic.

Rendering

TruckHub draws things in two different places, and for each one you can choose whether your **graphics card (GPU)** or your **processor (CPU)** does the drawing. The default for both is the GPU, which is right for almost every PC. These switches exist for one situation: your *game* is already pushing your graphics card very hard, and TruckHub's own drawing is getting squeezed.

> **IMPORTANT** **Both switches are off by default. If TruckHub looks and feels fine, leave them off.** A change only takes effect the next time TruckHub starts - Settings shows "Restart TruckHub to apply your change" with a **Restart now** button the moment you flip one. Neither switch affects how the game itself looks or runs in any direct way, neither can lose any data, and turning it back off and restarting undoes it.

Draw the HUD windows on the CPU

Controls: TruckHub's own windows - the main dashboard, the EXT gauges window, the job log window and Settings. It does *not* include the GPS map, which has its own switch below.

What it does: normally Windows hands the drawing of these windows to your graphics card. With this on, your processor does that work instead, so TruckHub stops using the graphics card at all (it showed 0 MB of graphics memory in testing) and can no longer compete with the game for it.

Turn it on if: while your game is running flat out, the HUD stutters, freezes for several seconds, or shows black or see-through boxes where gauges should be - even though the game itself is smooth. That pattern means the HUD is waiting its turn behind the game for the graphics card. It is most likely with very high game resolutions, heavy image scaling or upscaling, maxed-out graphics settings, or a graphics card whose memory is nearly full.

What it costs: a small amount of processor time. With nothing happening, TruckHub used about 0.6% of one processor core in our testing. While you drive it uses more, because the gauges are animating; to see what it costs on *your* PC, open Task Manager's Details tab and watch the CPU column for TruckHub.exe with the switch on and off. On a slow processor, or in a game that is already limited by your processor rather than your graphics card, the extra work could cost the game a little speed - so if you turn this on and the *game* gets worse, turn it back off.

Leave it off if: nothing is wrong. There is no benefit to switching it on "just in case".

Draw the GPS map on the CPU

Controls: only the GPS map window (and the map shared over your network if you use LAN Mode). The map is not drawn by TruckHub's own windows at all - it runs inside a built-in web browser (Microsoft Edge WebView2) with its own graphics acceleration, which is why it needs its own switch.

What it does: it turns that browser's graphics acceleration off, so your processor draws the map instead of your graphics card. The map looks the same and held the same frame rate in testing - about 30 frames per second while following a moving truck, on both settings.

Turn it on if: the game slows down or stutters specifically while the GPS map is open and your graphics card is already near its limit - the map is then competing with the game for the graphics card.

What it costs: processor time. Following a moving truck, the map used about 13% of one processor core with this on. With the graphics card doing the drawing, the browser still used between roughly 5% and 28% of a core for its own work in repeated tests, plus the graphics card itself - that GPU-mode figure varied a lot from run to run, so don't read the two numbers as a precise comparison. The map is also somewhat slower to load with this on (about 3.5 s of processor time versus 1.8 s). These figures come from the developer's PC (a Ryzen 7 5800X3D with an RTX 4070 graphics card), a fast desktop processor; **a slower processor will spend a larger share of itself on the map**.

Leave it off if: your game is limited by your processor (its CPU usage is high while its graphics card usage is not), or you have a slower processor. Moving the map onto a processor that is already busy makes things worse, not better.

Which one should I try?

Everything is smooth	**Leave both off**
The HUD stutters, freezes for seconds, or shows black / see-through boxes while the game is demanding	**Draw the HUD windows on the CPU**
The game stutters mainly when the GPS map is open, and your graphics card is near 100%	**Draw the GPS map on the CPU**
Both of the above	**Both - they're independent**
The game gets *worse* after turning one on	**Turn it back off - your processor is the limit, not your graphics card**

Change **one** switch at a time and drive for a while before deciding - the two fix different problems, and changing both at once makes it impossible to tell which one helped. TruckHub's log records what it is using at every start, for example *Render mode: HUD windows on CPU (software) [setting], GPS map on GPU [setting]* (see Section 10 for where the log lives). If the `TRUCKHUB_RENDER` environment variable is set on your PC (to `software` or `hardware`) it overrides both switches, and Settings says so.

UI Setup

Units

Three independent toggle buttons: speed (**MPH** / **KM/H**), cargo weight (**T** for tonnes / **LBS**) and fuel & AdBlue volume (**GAL** / **L**). Press one to flip it. The dashboard's speed, distances, weights and fuel economy (MPG or L/100km) all follow them, and the Job Log's mileage follows the speed toggle. Mix them freely - miles with litres is allowed.

Display components

A full on/off checklist for every component on the main dashboard: current speed, odometer, speed limit sign, in-game clock, cruise control, parking brake, lights status (beams / sidelights / beacon), gear + RPM, route (source / destination), cargo + weight, payout, distance, time left, fuel, trip fuel forecast (Trip Needs), fuel economy, AdBlue, and the warning lights. Turning something off shrinks the card to fit - nothing leaves a blank gap behind. The EXT window has its own, separate checklist (Section 05).

Speed alerts

Two optional sounds that save you glancing at the speed. Each has its **own on/off checkbox, its own volume slider (0-100%, default 50%) and its own Test button**, so you can keep one and silence the other. Both are on by default.

Speed limit change beep - one short beep whenever the posted speed limit changes. A **higher** tone means the limit went up, a **lower** tone means it went down. It stays silent when you switch between mph and km/h, and when limit information first appears after the game loads, and it sounds at most once every three seconds so a run of quick changes through a town gives one beep rather than a stutter. Because it can fire often, it is the one most people will want to turn down.

Over-limit warning - a short double beep the moment your speed readout turns **red** (more than 5 over the limit - the same point the speed number changes from amber to red). It sounds once per crossing, not continuously, and not when TruckHub first connects or when you un-pause already over the limit. If the limit drops and puts you over it, you hear this warning only, not both sounds.

The sounds are generated by TruckHub itself and play through your normal Windows audio output. The **Test** button on each row plays that sound at the volume you have set (the limit beep's Test plays the "up" tone and then the "down" tone).

Fatigue simulation

A single checkbox telling TruckHub whether you've enabled the game's own fatigue simulation. The game reports the same countdown value whether the simulation is on or simply hasn't started counting yet, so TruckHub can't detect this on its own; the Rest timer row only appears once you've confirmed it here.

SECTION 08

Mod Load Order

Opened with the Mod Order button on the main dashboard's title bar. TruckHub scans your installed mods for each game and suggests an order to load them in, based on community-recommended category tiers (map mods, vehicle mods, sound mods, and so on, each in the order that tends to cause the fewest conflicts).

Each game has its own **Scan** button, and a scan only happens when you press it - TruckHub never scans your mods on its own. It reads every mod's own manifest and, where that isn't specific enough, falls back on its file name. A mod and its "Open Def" companion always stay together. The groups, in loading order from the top:

1	Map Mods (loads first)
2	UI & Menu Layouts
3	Economy & Progression
4	Sound Frameworks
5	Graphics & Season Weather
6	Physics Systems
7	Interior & Cabin Accessories (including lights and tuning packs)
8	Standalone Trucks
9	Paint Jobs & Skins
10	Wheel & Tire Packs
11	Trailers & Cargo Add-ons
-	Unsorted - place manually. Anything TruckHub isn't confident about goes here instead of being guessed into a group.

This is a reference list only. TruckHub never reorders, enables, or disables anything in your Mod Manager - you read the suggested order here and apply it yourself. If your mod folder isn't in the usual place, set it under *Mod folder locations* in Settings (Section 07).

IMPORTANT This is the community's best-effort order, not a guarantee. Some mods are fussy about load order regardless of category, and can still conflict or crash even when correctly ordered. Treat it as a strong starting point, not a fix for every problem.

SECTION 09

Quick Reference

The two hotkeys

`Ctrl + Alt + G`	Capture the current H-shifter slot for gearbox calibration (Section 07). Must be pressed while the game window is focused and you're actually in the gear you want to label.
`Ctrl + Alt + F8`	Reroute the GPS map from where the truck is now (Section 04). Works from inside the game; nothing in ATS or ETS2 is bound to F8, so it can't also trigger a game action.

Where things live

TruckHub keeps its own files in `%APPDATA%\TruckHub-Dev` (paste that into the Windows Explorer address bar): your settings, the remembered gearbox calibrations, the remembered total of the job in progress, the LAN Mode certificate, and a `logs` folder holding `truckhub.log`. The unpacked GPS map files and the map browser's working data are under `%LOCALAPPDATA%\TruckHub-Dev`. The plugin it installs lives in each game's `bin\win_x64\plugins` folder. TruckHub writes nowhere else (the Mod Order tool only *reads* your mod folder).

Color legend, condensed

Green	Normal, safe, or a driving aid working as intended (cruise, retarder, engine brake, turn signals, low beam, sidelights).
Amber	Needs attention soon, or a deliberate state that changes handling (diff lock, lift axle up, low fuel/AdBlue, beacon, near-due rest timer).
Red	Safety-critical fault or danger: low air pressure, oil pressure, overheating, dead battery/charging failure, over the speed limit by more than 5, an axle unexpectedly off the ground.
Blue	High beam, specifically - the one near-universal dashboard exception to the green/amber/red scheme.
Dim grey	Off, disengaged, or not applicable right now.
White	A neutral reading with no state to flag - a plain number or label.

TruckHub reads game telemetry only - it cannot see or modify anything else on your system. The only thing it ever sends anywhere is its update check, which just asks GitHub whether a newer version exists; LAN Mode, if you turn it on, shares the map on your own home network only. If something on the HUD ever looks wrong, the game's own dashboard is always the source of truth: TruckHub is a second window onto the same numbers, not a separate calculation.

SECTION 10

Troubleshooting

Symptom first, then what to do about it.

The card only says *Waiting for ETS2 / ATS to start…* although the game is running	TruckHub's plugin isn't loaded by the game. **1.** If TruckHub installed the plugin while the game was already open, quit and restart the game - plugins are only loaded at game start. **2.** Check that `scs_telemetry_hub.dll` is in the game's `bin\win_x64\plugins` folder; if it isn't, TruckHub didn't find the game, so set the folder under *Game install locations* in Settings and restart TruckHub. **3.** The plugin uses a local-only network port; if another program already holds it, the plugin says so in the game's own log (`game.log.txt` in the game's Documents folder).
The card says *Paused* or *No active job*	Both are normal. *Paused* means the game is paused; *No active job* means just that - the job panel appears when you take a job.
The speed limit sign shows a grey --	Either the current road has no limit data, or the in-game option is off: turn on **Options > Gameplay > HUD > Route Advisor: speed limit** (Section 01).
The gear box shows something like S4 instead of a gear number	That's an H-shifter gearbox TruckHub hasn't been calibrated for yet. It shows the raw shifter slot rather than a confidently wrong gear. See Gearbox calibration in Section 07 (Ctrl+Alt+G).
The REST row is missing	It only appears once you tick *I have fatigue simulation enabled in-game* in Settings > UI Setup (Section 07).
There is no route on the GPS map	The route appears once a job's destination is known. With no job, press **Pin** and click the map to route somewhere (Section 04). TruckHub's GPS is separate from the in-game one, so the two can disagree about which road to take.
The map is slow to appear after installing or updating TruckHub	It is unpacking its map files in the background - a few minutes of heavy processor use, once. See Section 04.
A route stays where it was after I pulled off for fuel	That's deliberate: nothing is rebuilt until you are a mile or more off the route. Press **Reroute** (or **Ctrl+Alt+F8**) when you want a new one, or turn on automatic rerouting in Settings (Section 07).
The Distance box says "of" a different total than I expected	The total is the job's route length, remembered while the job is in progress. A job TruckHub first sees part-way through falls back to the job's quoted length (Section 02).
The HUD stutters, freezes, or shows black or see-through boxes while the game is demanding	Try **Draw the HUD windows on the CPU** in Settings > App Configure > Rendering (Section 07). Restart TruckHub to apply it.
The game stutters while the GPS map is open	If your graphics card is near its limit, try **Draw the GPS map on the CPU** (Section 07). If your processor is the bottleneck instead, leave it off, or simply close the map window when you don't need it.

TruckHub's memory use looks high with the GPS map open	Some of that is expected: to route over a whole continent, TruckHub has to hold its road network in memory, and the map itself runs in a browser process of its own. The route finder stores the network compactly (roughly 50 MB for the American map and 62 MB for the European one, about a fifth of what earlier versions needed).
A telemetry or other plugin crashes the game when it exits	TruckHub installs one plugin, designed to shut down cleanly when the game closes. Every extra plugin in the plugins folder is one more module that can fault at exit, and a crash report that names a different `.dll` points at that plugin, not TruckHub. If the report names `scs_telemetry_hub.dll`, please send it along with the log (below).

Reporting a problem

Open `%APPDATA%\TruckHub-Dev\logs\truckhub.log`. It is a plain-text event log - connects and disconnects, job start and finish, the render mode in use, plugin install results and any errors - with the date and time on every line. Attach it, together with the game's own `game.log.txt` and `game.crash.txt` if the game crashed, to an issue at github.com/UncappingBadger/Badgers-Truckhub-for-ATS-and-ETS2 and say what you were doing at the time.